

BLAST FURNACE PRODUCTION SECTION												ON-DT		ON-Month	
DATE: 1. Mar. 2025										Crude steel To HM Ratio		0.961		0.961	
DAILY PARAMETERS REPORT										STOCKS		ON-DT		Prev.Day	
Monthly Plan (T)		ABP 599000		OPP 525000		Sinter Prod. / Nut Coke		EB 3100		Sinter stock		High lines (1-7)		1400	
Daily Rate (T/Day)		18407				SP - II 8325		51		I/Ore lump stock		Em.Storage		11500	
H.M.SENT TO SMS's		18407		COAL STOCK		277870 M/c-II		6157				OHP # 2		9401	
				Dry Coke ==>		7509 TOTAL		23372		51		BF		65908	
												CO		20511	
												Bfs		760	
												FINE COAL		758	
												RAW COAL		350	
												SHIPPING REPORT		HM (PCM)	
														PROD.	
														0.00	
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HOT METAL AND SLAG QUALITY

Parameter	BF-1	BF-4	BF-5	BF-6	BF-7	BF-8	SHOP	TILL	(BF-4 to 7)
Avg. % 'Si'		0.69		0.76	0.58	0.60	0.63	0.63	0.42
Min/Max		0.53/0.91		0.42/1.03	0.34/0.83	0.45/0.75			
Till % 'Si'		0.69	#DIV/0!	0.76	0.58	0.60			0.62
Avg. % 'S'		0.027		0.029	0.030	0.027	0.028	0.028	
Min/Max		0.020/0.034		0.020/0.047	0.020/0.038	0.020/0.043			
Avg. % MgO		7.83		7.46	7.26	8.05	6.12		
Min/Max		7.53/8.12		7.09/7.86	6.82/7.64	6.93/8.67			
Avg. % Al2O3		17.23		17.74	17.98	20.45	14.68		
Avg. % FeO		0.7		0.69	0.93	0.82	0.63		
Avg. % K ₂ O		0.39		0.43	0.49	0.47	0.36		
BASICITY(-)		1.02		0.99	1.01	0.98	0.80		
Min/Max		0.97/1.09		0.95/1.02	0.94/1.04	0.93/1.03			
Avg. % 'P'		0.16		0.16	0.16	0.10	0.15		
Min/Max		0.15/0.18		0.12/0.16	0.14/0.17	0.09/0.13			

Fce.No.	Si→1.00	Si→0.85	No.of Casts	Tap to Tap
4	0	2	8	0
5			0	
6	1	4	8	0
7	0	0	10	0
SHOP	1	6	26	0

H ₂ IN TOP GAS						
	BF-1	BF-4	BF-5	BF-6	BF-7	BF-8
G		3.70	STOP	5.20	4.60	

<u>CO/CO₂ RATIO</u>						
G		1.28		1.27	1.33	
<u>% Gas Util.</u>						
G		43.87		43.98	42.93	47.53

MIXER	%Si	%Mn	%Sul	%P
Mixer-III				
Mixer-IV				

PELLET CHEMICAL AND SEIVE ANALYSIS

	Fe	SiO ₂	Al ₂ O ₃	CaO	MgO	+ 10 MM	- 5 MM	M.Size	Basicity	Slag.Comp.
BF # 8 =>	62.92	4.47	3.29	1.38	0.41	57.70	1.60	11.10	0.31	9.55
BF # 7 =>	62.45	4.79	3.46	1.38	0.57				0.29	10.20
BF # 5 =>									#DIV/0!	0.00
BF # 4 =>									#DIV/0!	0.00

SKIP IRON ORE				
	Fe (%)	SiO ₂ (%)	Al ₂ O ₃ (%)	Moist.
BF # 4	64.24	3.92	1.40	2.8
BF # 5	STOP			
BF # 6	63.00	5.56	1.64	4.8
BF # 7	64.22	3.73	1.70	4.2
BF # 8	61.52	4.92	3.14	4.6
Avg.	63.25	3.63	1.58	4.1

BF SKIP /ORE SEIVE ANALYSIS			
Fce.No.	+40 mm	- 10 mm	M.Size
BF # 4	28.70	4.87	44.03
BF # 5			
BF # 6	30.80	4.77	44.13
BF # 7	29.43	3.73	45.03
BF # 8	33.93	5.78	37.98
AVG.	30.72	4.79	42.80

% FINES IN BF SKIP SINTER				TOTAL Fe				% FINES IN BF SKIP COKE			
Fce.No.	-10 mm	- 5 mm	M.Size	SHIFT A	SHIFT B	SHIFT C		Fce.No.	-40 mm	- 25 mm	M.Size
BF # 4	31.20	6.73	14.50	52.73	52.45	51.86		BF # 4	24.13	8.93	35.13
BF # 5								BF # 5			
BF # 6	34.07	7.90	14.80	50.73	52.99	52.62		BF # 6	26.30	7.83	35.80
BF # 7	27.53	6.57	15.90	50.28	49.76	50.55		BF # 7	23.63	8.70	35.37
BF # 8	29.57	3.18	15.88	51.78	52.25	52.60		BF # 8	28.17	9.90	32.48
AVG.	30.59	6.10	15.27					AVG.	25.56	8.84	34.70

COKE QUALITY									
PROXIMATE ANALYSIS					COLD STRENGTH			HOT STRENGTH	
	Moisture	Ash	V.M	F.C	M-40	M-10	Sulphur	CSR	CRI
CSP-I: A	3.3	15.0	0.9	83.6			0.59		
CSP-II: A	4.3	14.9	0.9	83.8	81.6	7.4	0.63	64.6	23.8
CSP-III: A	5.0	14.9	0.8	83.8			0.60		
CSP-IV: A	0.3	14.8	1.0	83.8	87.4	6.2	0.56		
BF # 8 STOCK HOUSE SURFACE MIX.COKE =>								65.4	23.5

COAL BLEND (%)													
	DATE	PCC	MCC	Aus(H)	BENGA (H)	LVHR (H)	TUHP(hard)	Russian (S)	AUS(S)	USA(H)	USA(S)	MABU (H)	% IMP
CHP - I	13-Feb-25	13.0	3.0	39.0	0.0	10.0	0.0	8.0	14.0	13.0	0.0	0	84.0
CPP - II	24-Feb-25	12.0	3.0	30.0	7.0	9.0	0.0	7.0	10.0	5.0	5.0	12	85.0

CHEMICAL ANALYSIS

SINTER PLANT-2											
NORM(Basicity)						6 Kg/T of Sinter					
NORM(Mgo)						25 Kg/T of Sinter					
(1.70 - 1.75) (13-01-25) Nut Coke						(2.75 - 3.25) (05-07-24) MILL SCALE					
SAMPLE	% Fe	% FeO	% SiO ₂	% Al ₂ O ₃	% CaO	% MgO	% MnO	CaO-SiO ₂	NORM	Basicity	Slag Component
AO	50.59	9.40	8.01	3.04	13.72	3.16	1.71	5.71	IN	1.71	27.82
A1	50.38	9.40	8.06	2.71	14.08	3.09	1.75	6.02	IN	1.75	C I (SP # II)
A2	51.09	9.40	7.75	2.68	13.38	3.21	1.73	5.63	IN	1.73	Coke
BO	50.16	9.42	8.29	2.66	14.33	3.11	1.73	6.04	IN	1.73	Flux
B1	50.41	9.42	8.31	2.22	14.33	3.33	1.72	6.02	IN	1.72	DTI
B2	51.09	9.42	7.57	2.93	13.01	3.11	1.72	5.44	IN	1.72	
CO	50.07	9.42	8.38	2.46	14.49	3.10	1.73	6.11	IN	1.73	
C1	50.50	9.42	8.07	2.38	14.26	3.06	1.77	6.19	OUT	1.77	
C2	50.31	9.42	7.96	2.82	14.00	3.28	1.76	6.04	OUT	1.76	
DAY AVG.	50.51	9.41	8.04	2.66	13.96	3.16	1.74	5.91	IN	1.73	

SINTER PLANT-3											
NORM(Basicity)						0 Kg/T of Sinter					
NORM(Mgo)						12 Kg/T of Sinter					
(1.70 - 1.75) (13-01-25) Nut Coke						(2.75 - 3.25) (05-07-24) MILL SCALE					
SAMPLE	% Fe	% FeO	% SiO ₂	% Al ₂ O ₃	% CaO	% MgO	% MnO	CaO-SiO ₂	NORM	Basicity	Slag Component
AO	52.98	9.22	7.09	2.62	12.00	2.92	1.69	4.91	OUT	1.69	26.04
A1	51.48	9.22	7.84	2.66	13.15	3.09	1.68	5.31	OUT	1.68	DTI
A2	51.50	9.22	7.82	2.74	13.07	3.11	1.67	5.25	OUT	1.67	
BO	52.51	9.07	7.05	2.99	12.14	3.09	1.72	5.09	IN	1.72	
B1	52.05	9.07	7.39	2.91	12.49	3.04	1.69	5.10	OUT	1.69	
B2	51.97	9.07	7.57	2.40	13.16	2.80	1.74	5.59	IN	1.74	
CO	51.53	9.07	7.45	2.99	12.50	2.91	1.68	5.05	OUT	1.68	
C1	51.48	9.07	7.73	2.62	12.93	2.94	1.67	5.20	OUT	1.67	
C2	51.20	9.07	7.54	3.30	12.94	3.33	1.72	5.40	IN	1.72	
DAY AVG.	51.86	9.12	7.50	2.80	12.71	3.03	1.70	5.21	OUT	1.70	

Sample ID	% Fe	% FeO	% SiO ₂	% Al ₂ O ₃	% CaO	% MgO	% MnO	CaO-SiO ₂	NORM	Basicity	Slag Comp
SP-3/S/MC/2A0	52.09	9.81	7.50	2.75	12.72	2.92	1.70	5.22	OUT	1.70	51.36
SP-3/S/MC/2A1	51.39	9.81	7.90	2.65	13.19	2.99	1.67	5.29	OUT	1.67	26.23
SP-3/S/MC/2A2	51.29	9.81	7.91	2.70	13.27	3.16	1.68	5.36	OUT	1.68	26.43
SP-3/S/MC/2B0	51.32	9.69	7.79	2.98	13.23	3.05	1.70	5.44	OUT	1.70	C I (SP # III)
SP-3/S/MC/2B1									#DIV/0!	#DIV/0!	Coke
SP-3/S/MC/2B2									#DIV/0!	#DIV/0!	Flux
SP-3/S/MC/2C0									#DIV/0!	#DIV/0!	DTI
SP-3/S/MC/2C1	51.22	9.69	7.88	3.33	13.19	2.87	1.67	5.31	OUT	1.67	
SP-3/S/MC/2C2	52.88	9.69	6.87	2.78	11.78	3.15	1.71	4.91	IN	1.71	
DAY AVG.	51.70	9.75	7.64	2.87	12.90	3.02	1.69	5.26	OUT	1.69	

BLAST FURNACE DEPARTMENT

Stoppage, L.Blast, Reduced Blast & offrod Duration and Reasons for Dated

1-3-25

on Dtd 2-Mar-2025

8:43

F.N	Duration		REASONS	ST	LB	RB	FB	Down Time	Furnace No	1	4	5	6	7	8	TOTAL
I			== FCE. BLOWN OUT ==						ABP	0	77000	77000	79000	119000	247000	599000
									OPP	0	77250	0	80750	119700	247300	525000
IV	04:20	06:00	LT. SINTER FINES FEEDER MOTOR CHANGING						M/Rate	0	79608	0	86986	140120	263903	570617
									Weekly Plan	0	15050	0	16650	21750	49100	102550
							24:00:00	0:00:00	Prod.(Cumm.)	0	15800	0	17072	26485	50262	109619
V	06:00	05:59		23:59					Fulfillment	#DIV/0!	105.0	#DIV/0!	102.5	121.8	102.4	106.9
				23:59			0:01:00	23:59:00	Today'sPlan	0	2550	0	2650	3950	8100	17250
VI	11:30	11:50	RT. COKE LSS U/ELECT. FOR CARBON BUSH CHANGING						HM Prod	0	2568	0	2806	4520	8513	18407
			LEVEL DOWN 3Mtr. , OXY REDUCE						Cu.Prod	0	2568	0	2806	4520	8513	18407
			ST-21 GSSV OPENING PROB. ('B' SHIFT)						D/Rate	0	2568	0	2806	4520	8513	18407
			LT. SINTER GATE CLOSING PROB., CROSS CHARGING (C SHIFT)						No.OF SKIPS (6:00 TO 6:00)						Charge	Total
	03:55	04:10	RT. SLI & LT. RADAR BOTH NOT WORKING						A		196		221	282	46	699
	04:00	06:00	MOTOR & GEAR BOX CHANGED			2:00			B		205		218	278	48	701
						2:00	22:00:00	0:30:00	C		203		192	280	44	675
VII	14:00	14:10	LC OFF						Total skips	0	604	0	631	840	138	2075
									Stoppage on Date			23:59				23:59:00
							24:00:00	0:00:00	Low Blast on Dt							0:00:00
VIII			==NIL==						Reduced blast on Dt				2:00			2:00:00
							24:00:00	0:00:00	No of Tappings	0	8	0	8	10	12	38
									No of Not Dry casts	0	0	0	0		0	0
			TOTAL	23:59:00		2:00:00	94:01:00	24:29:00	Coke Rate On Date		447		433	421	381	408
									Coke Rt for month		447		433	421	381	408
									Hot Blast Temp.		1002		1000	1004.917	1100	1043

COAL DUST INJL (Kg/THM)

BF		CDI	HM Production on	
No.	DAILY/ Date	Rate(kg/thm	that date(T)	CDI in tonnage(T)
1	25.03.2021	124	1852	230.15
4	14.12.2022	226	1250	281.90
5	03.11.2024	209	1118	233.62
6	10.10.2022	206	1821	313.14
7	07.08.2024	158	3083	488.15
8	05.11.22	192	7651	1472.00
SHOP	21.10.24	160	14750	2360.00

CDI On Tonnage Basis

BF		Best CDI On tonnage	HM Production on
No.	DAILY	Basis(T)	that date(T)
1	25.03.2021	230.15	1852
4	18.10.2024	410.99	1954
5	07.11.2024	427.53	2304
6	19.09.2022	334.99	2044
7	10.12.2022	540.30	4176
8	30.11.2022	1575.97	8793
SHOP	25.01.2023	2721.79	18771

Annual ABP Target(FY 2024-25)		Till Date
HM Target(MT)	6.60	5.36
Coke RATE(kg/thm)	429	394
CDI Rate(kg/thm)	133	135
FUEL RATE(kg/thm)	588	551
Financial Data		
Cost of HM Production in FY 2023-24 (per ton of HM)	Rs 35524	
Cost of HM Production in FY 24-25 till Oct'24(per ton of HM)	Rs 33554	
Cost of HM Production in Oct'24(per ton of HM)	Rs 32291	
Contribution Margin per ton (Profit) for Pig Iron in oct-24	Rs 8995	
Cost of Skip coke in FY 24-25 Till Oct'24 (per ton)	Rs 38175	
Cost of Skip Coke in Oct'24(per ton)	Rs 36476	
Previous Year ABP vs Actual (FY2023-24)	ABP	ACTUAL
HOT METAL(MT)	6.20	5.99
COKE RATE(kg/thm)	430	455
CDI RATE(kg/thm)	125	120
Fuel Rate(kg/thm)	587	597
Pig Iron Produced in FY 2023-24	67695 T	
Pig Iron Produced in FY 2024-25	26673 T	

Metal ladle position				Slag ladle position			
In circulation	39	+80	1	Total	In Use	Under	Under
Open Hearth		+75	2			M/Maint.	E/Maint.
Reserve	23	+70	18	32	32		
U/Repair	3	-70	13	Total No of Hot Metal loads			128.0
Cooling	6	Total	34	No.of Tarpedo			30.0
Mixer Position & M- III				% Screen Loss		Loads sent to OH	0
Pouring AT M-IV						Loads sent to CS	128
6:00:00 AM SMS-III (TPD)						Loads sent to PCM	0
						Loads sent to FDY	0
%SINTER FINES	SP # II		SP # III	% COKE FINES		BF#7	BF#8
	13.98		21.47			0.00	8.85
Coal Dust Injection	BF#1	BF#4	BF#5	BF#6	BF#7	BF#8	SHOP
Coal Consum.(T)	0.00	291.85	0.00	346.49	501.76	1216.00	2356.100
Coal Rate KG/THM	#DIV/0!	114	#DIV/0!	123	111	143	128.000
Coal Prog. KG/THM	0	16	0	19	27	66	128
Raw Mat. Position							
(FROM RMD)	L/STONE	Mn.Ore	Qz.	Conv.slagn	CDI COAL	Pellet (OHP)	OHP B
Receipt	0	0	0	0	1265	2663	1366
Stock	13471	38	3211	1500	1387	16266	33791

RECORDS

B.Fces.	Hot Metal Production		COKE RATE		CDI RATE	
	MONTHLY		MONTHLY		MONTHLY	
BF # 1	61574	AUG.'17	436	MAR.'12	124	JAN.'24
BF # 4	82493	MAY.'24	438	MAR.'23	154	OCT.'24
BF # 5	82758	MAR.'22	418	FEB.'09	162	OCT.'24
BF # 6	89535	JAN.'09	397	JAN.'02	144	OCT.'22
BF # 7	128247	OCT.'17	428	MAR.'19	124	NOV.'22
BF # 8	252731	DEC.'24	370	SEP.'21	160	DEC.'23
SHOP	555571	DEC.'24	414	SEP.'21	150	OCT.'24
PCM POURING	41	2451	30-06-2023			